

Technical Summary: Evans' Index Plus Calculator

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Live Application URL: <https://rasan17.github.io/evans-index-plus/>

Part 1: How the Global Counter Engine & Separate Metrics Were Solved

1. The Challenge with Local vs. Global Counting

In initial versions, browsers were displaying separate counts per computer because:

- Free public counter endpoints returned `HTTP 410` (deprecated API versions) or failed CORS preflight checks (`OPTIONS` HTTP headers).
- Whenever a network call failed, the browser silently fell back to `localStorage` (which only stores data locally on that specific computer/browser).

2. The Solution: CounterAPI v2 Engine & Cache-Busting

- **Direct REST Querying:** Switched to **CounterAPI v2** using direct URL queries without custom preflight headers.
- **Cache-Busting (`?_t=TIMESTAMP`):** Appended live Unix timestamps (`Date.now()`) and `{ cache: 'no-store' }` to every HTTP request. This prevents Chrome, Safari, and mobile browsers from serving stale cached numbers and forces them to fetch the live database total every time.

3. Separating "App Views" vs. "Calculations Done"

To ensure **App Views** and **Calculations Done** operate as two completely independent global counters:

- **App Views → Channel 1 (`up_count`):** Increments globally whenever any visitor worldwide opens or refreshes the web application. Endpoint: <https://api.counterapi.dev/v2/test/test/up>
- **Calculations Done → Channel 2 (`down_count`):** Reads existing totals on page load *without incrementing*, and increments globally **ONLY** when a user enters valid cranial

measurements and completes an Evans' Index calculation. Endpoint:

<https://api.counterapi.dev/v2/test/test/down>

Part 2: How the Longitudinal Bar Graph Was Engineered to Work Without Dates

1. The Problem: The NaN Sorting Bug

Originally, the application:

1. Filtered time points using `tp.date && tp.index !== null`. If a user entered measurements without selecting a calendar date, the row was completely ignored.
2. Attempted to sort undated rows using `a.number - b.number`. Because `number` was an undefined property on row data objects, `undefined - undefined` evaluated to `NaN` (Not-a-Number) in JavaScript, causing `Array.prototype.sort()` to fail silently and suppress the Chart.js rendering pipeline.

2. The Solution: Date-Independent Filtering & Fallback Sorting

- **Date-Independent Qualification:** Updated the filtering logic to qualify any row with valid numerical measurements, regardless of whether a date is provided:

```
const valid = timePoints.filter(tp => tp.index !== null &&
  !isNaN(tp.index));
```

- **Fail-Safe Chronological Sorting:** Updated the sorting algorithm to order chronologically when dates are present, while safely preserving insertion order (`return 0`) when dates are omitted—eliminating `NaN` sorting errors completely:

```
valid.sort((a, b) => {
  if (a.date && b.date) return new Date(a.date) - new Date(b.date);
  if (a.date) return -1;
  if (b.date) return 1;
  return 0;
});
```

3. Dynamic Smart Bar & Tooltip Labeling

To ensure the graph displays informative X-axis labels under all user input scenarios, the renderer evaluates a 4-tier fallback hierarchy:

Input Provided	X-Axis Bar Label Example	Tooltip Title Example
Context + Date	Pre-Op (Jan 15, 2026)	Pre-Op (Jan 15, 2026)
Context Only	Pre-Op	Pre-Op
Date Only	Jan 15, 2026	Scan Date: Jan 15, 2026
Neither (Undated)	Scan #1 , Scan #2	Scan #1

 Repository Files Summary

File Name	Purpose
index.html	Structural HTML5 markup, patient details, longitudinal measurement table, KPI dashboard, printable clinical report layout, and copyright notices.
style.css	Glassmorphism dark aesthetic, responsive table design, threshold badges, counter stat bar, and @media print print styling.
script.js	Real-time calculation engine, CounterAPI v2 global counter engine, Chart.js bar renderer, and sample case preset loader.
evans_diagram.jpg	High-definition anatomical CT/MRI scan diagram showing exact Line A (widest frontal horn tips) and Line B (maximal inner skull diameter).